A Two-Layer Evaluation of Speech-to-Text and LLM-Based Terminology Extraction Systems for Multilingual Education

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***Abstract*—Personalized instruction can substantially improve learning outcomes, but it is hard to scale, particularly for spoken lecture and video content, where the material must first be understood before it can be adapted to a learner. This study presents a two-layer experimental *evaluation* of a speech-based learning support system designed to improve the accessibility of multilingual educational content for beginner-level users. Rather than proposing a new architecture, the work is a systematic benchmark of how off-the-shelf automatic speech recognition (ASR) and large language model (LLM) components behave *jointly* in a multilingual educational setting. In the first layer, multiple ASR systems are comparatively evaluated in terms of transcription accuracy (WER, CER) and computational efficiency (latency, real-time factor). In the second layer, the resulting transcriptions are processed by two LLMs to perform domain-specific terminology extraction and context-aware explanation generation. The semantic outputs are evaluated across Turkish and English datasets using accuracy, coverage, and linguistic-suitability metrics. Experimental results show that Whisper-based ASR models provide more reliable transcriptions for downstream processing, while GPT-4.1 achieves higher terminology coverage and explanation accuracy than the lightweight alternative. The findings highlight the strong dependency between transcription quality and semantic-processing performance, and emphasize the importance of jointly considering ASR and LLM components in speech-driven educational systems. We additionally formalize the evaluation protocol, report reproducibility details, and discuss the limitations of the study in a dedicated section.**

***Index Terms*—Automatic Speech Recognition, Terminology Extraction, Large Language Models, Multilingual Educational Content, Evaluation Benchmark**

I. Introduction

With the recent growth of digital technologies, educational content is increasingly delivered through audio- and video-based media. Lectures, instructional videos, and multimedia material are widely used to convey knowledge across many domains. However, such content is often prepared under the assumption that all users share a similar level of prior knowledge. This assumption limits comprehensibility, accessibility, and learning efficiency for beginner-level users, particularly in segments that contain domain-specific terminology [1], [2].

Tailoring instruction to the individual learner is one of the most effective ways to close such gaps: in his classic study, Bloom reported that one-to-one tutoring with mastery learning raised the average learner's performance by roughly two standard deviations over conventional classroom instruction [3]. However, human one-to-one tutoring is too costly to scale to large populations, which motivates automated approaches to personalization. A long line of work has sought to approximate the benefits of individualized instruction at scale, including intelligent tutoring systems and adaptive educational/hypermedia systems [4], and learner-modeling techniques such as knowledge tracing that estimate what a student already knows in order to adapt the material accordingly [5]. Meta-analyses indicate that such systems can be nearly as effective as human tutors [6], and more recently large language models have been explored as a flexible substrate for instructional support, personalization, and content generation [2]. Across these approaches, an adaptive learning support system typically comprises three stages: (i) ingesting and understanding the source content, (ii) modeling the learner's needs and prior knowledge, and (iii) adapting the delivery, for example by generating learner-appropriate explanations. For spoken lecture and video material, however, this pipeline cannot begin with stage (ii) or (iii): the content is locked in the audio signal, so the first stage must convert speech to text and then provide term-level semantic support on top of it. This first stage is the focus of the present study.

Accordingly, the first stage—making audio-based educational content comprehensible—requires that the speech signal first be converted to text. Manual subtitling and expert correction of large volumes of lecture content are costly and difficult to scale [7]. As a result, automatic speech recognition (ASR) has become a critical component in educational platforms, both for accessibility and for searchability. In recent years, transformer-based ASR approaches trained on large-scale multilingual data have reported more robust results under noise, accent variation, and domain shift (e.g., Whisper) [8]. Self-supervised approaches, in turn, aim to improve ASR accuracy by learning strong speech representations from

limited labeled data (e.g., wav2vec 2.0) [10]. Open-source speech-processing frameworks facilitate fair comparison of different ASR configurations and reproducible experimental setups (e.g., SpeechBrain) [11].

Transcription alone is not sufficient: for beginner-level users, the main difficulty is understanding the domain-specific terms (including multi-word terms) that occur in lectures, within their context. Prior work has reported that such terminological load limits comprehension and learning efficiency in educational settings, and that ontology- and term-based support mechanisms improve comprehensibility [1], [12]. This requirement motivates semantic processing steps such as automatic terminology extraction and explanation generation; recent work emphasizes the role of transformer-based approaches and systematic evaluation in terminology extraction [17]. Nevertheless, in the literature the quality of ASR output and the quality of LLM-based semantic output are often reported separately; studies that jointly evaluate the "ASR+LLM" pipeline in multilingual educational scenarios remain limited.

In this study, we conduct a two-layer evaluation that jointly measures (i) ASR accuracy and latency on audio extracted from Turkish and English educational videos, and (ii) LLM-based terminology extraction and context-aware explanation generation on the same transcriptions. When integrated as a plugin to a learning management system, such a system could support learning by presenting term–explanation cards next to the player, either during playback or after recording [20]. Our contributions are as follows:

- We compare multiple ASR approaches on multilingual (TR/EN) educational speech in terms of accuracy (WER/CER) and efficiency (RTF/latency), under off-the-shelf (zero-shot) configurations.
- We evaluate two LLM configurations on the same transcriptions in terms of terminology coverage and explanation accuracy.
- We report the effect of ASR–LLM component choices on the semantic outputs, recommend pipeline configurations for different application scenarios, and provide a formalized evaluation protocol together with reproducibility details.

II. Related Work

The spread of video-based learning in education has increased the importance of captions and transcripts for accessibility and learning support. However, the quality of automatic captioning can vary substantially depending on domain-specific terminology, speaker characteristics, and language/accent diversity. For this reason, evaluating ASR performance with a single averaged value is insufficient. Studies on higher-education lecture recordings show that the accuracy of different ASR systems varies with the data and speaking conditions [13]. Moreover, ASR errors are known to propagate negatively to downstream NLP tasks, which underlines the need for pipeline-level evaluation [7].

On the ASR side, Whisper stands out as a strong model trained on large-scale multilingual data [8], while wav2vec 2.0 offers high performance under low-resource labeled conditions through self-supervised learning [10]. Open-source tools such as SpeechBrain facilitate the comparison of different ASR systems [11]. Studies on low-resource languages such as Turkish also show that model choice is critical [14].

In the terminology extraction literature, rule-based and statistical methods were the dominant approach for a long time (e.g., the C-value/NC-value method for multi-word terms [12], and information-retrieval-based statistical measures [15]), after which deep-learning-based methods became widespread. Work on biomedical texts addresses term extraction with multiple measures [16], while the LiTeWi approach extends usage to the educational context by linking terms to ontological structures [1]. Recent work emphasizes the role of transformer-based approaches in terminology extraction [17].

Beyond transcription and term extraction, a broader line of research targets adapting instruction to the individual learner. Bloom's classic result that one-to-one mastery tutoring yields large gains over classroom instruction [3] motivated decades of work on scalable approximations, including intelligent tutoring systems and adaptive educational/hypermedia systems [4] and learner-modeling techniques such as knowledge tracing [5]; meta-analyses report that such systems can approach the effectiveness of human tutors [6]. More recently, large language models (LLMs) have been used as a flexible substrate for instructional support, personalization, and content generation [2], and controlled studies have shown that LLM-based tutoring with carefully designed prompting can improve engagement and learning outcomes while requiring minimal content preparation [18]. These systems generally operate in stages—understanding the content, modeling the learner, and adapting the delivery—where reliable understanding of the source content is a prerequisite for the later, learner-facing stages.

For spoken educational material, the content-understanding stage is itself a multi-step process in which speech is first transcribed by ASR and then processed semantically by an LLM. In the literature, however, the quality of ASR output and the quality of the downstream semantic output are typically reported in isolation: ASR work optimizes WER/CER [8], [10], while terminology-extraction work largely assumes clean text [12], [17]. Although ASR errors are known to propagate to downstream NLP tasks [7], studies that jointly evaluate the end-to-end ASR→LLM terminology pipeline—and especially across both a high-resource (English) and a lower-resource (Turkish) language—remain scarce. This study targets that gap by presenting a two-layer evaluation that measures how transcription quality and semantic-processing quality interact in a multilingual educational setting.

III. Method

A. System Architecture

The proposed system architecture consists of two main components: a speech recognition layer and a semantic analysis

TABLE I: Dataset used in the experiments (total ∼95 min).

ID	Title	Lang.	Domain	Dur.
TR-1	Project Management – Unit 7 Lecture 1	TR	Project Mgmt.	∼12 min
TR-2	Comprehensive Tourism Training for Tour Guides	TR	Tourism	12 min
TR-3	What Is Data Science? Intro to Data Science (Python)	TR	Data Science	∼14 min
ENG-1	Introduction to Agile Methodologies	EN	Project Mgmt.	∼13 min
ENG-2	What Is Sustainable Tourism?	EN	Tourism	∼9 min
ENG-3	Generative AI In A Nutshell	EN	AI	∼18 min
ENG-4	Understanding Thermal Radiation	EN	Heat Transfer	∼17 min

layer. In the first stage, audio recordings are converted to text using ASR systems. The transcriptions obtained at this stage form the fundamental input for all subsequent steps. In the second stage, the transcriptions are subjected to semantic analysis by providing an instruction (prompt) to LLMs. At this stage, domain-specific terminological expressions in the text are identified, and explanations understandable to beginner-level users are generated for these terms. The overall workflow of the system is shown in Fig. 1.



Fig. 1: System architecture. Eraser online tool was used for creating the figure.

B. Dataset

The dataset used in the experimental evaluations consists of video recordings representing Turkish and English educational content, and of audio extracted from these videos, as shown in Table I. The dataset comprises 7 videos and a total of approximately 95 minutes of audio. The content covers educational materials from different domains and includes examples in which domain-specific terminology is used in natural speech. All audio files were converted to 16 kHz sampling rate and mono-channel format to allow consistent experimental comparison. No manual cleaning or editing was applied to the audio data, so that the recording conditions were preserved. In addition, the selected video topics were chosen to have similar terminological density, so as to reduce the effect of domain differences on the results.

C. Speech Recognition Layer

The first stage of the system is the speech recognition layer, which converts the audio file to text. At this stage, multiple ASR systems with different architectural paradigms were evaluated. The goal is to select a configuration that balances accuracy and computational efficiency. Three ASR systems were used in the experiments:

- Whisper (Medium, Large-v3) – end-to-end transformer ASR
- Wav2Vec 2.0 – self-supervised transformer encoder
- SpeechBrain – hybrid ASR with language-model decoding

These models were selected among common open-source systems representing different ASR architectural paradigms. All models were evaluated on the same audio inputs with model parameters held fixed. This study intentionally adopts an *off-the-shelf (zero-shot)* evaluation design: no domain-specific fine-tuning was applied, and the default pre-trained weights were used. The rationale is to reflect realistic low-resource deployment conditions in which labeled data is scarce, and to benchmark the models "out of the box" without any adaptation.

D. Semantic Analysis Layer

The most suitable transcriptions obtained in the speech recognition layer are passed to the second stage of the system, the semantic analysis layer. Two LLMs were used in the semantic analysis process:

- GPT-4o-mini: an efficient baseline model in terms of low latency and computational cost,
- GPT-4.1: a reference model representing higher model capacity.

The semantic analysis process consists of two steps: (i) *terminology extraction*, identifying domain-related terminological concepts within the transcription; and (ii) *explanation generation*, producing context-aware explanations for the selected concepts that beginner-level users can understand.

In all experiments, the system instruction (prompt), the model parameters, and the generation settings were held fixed. The models were accessed through the OpenAI API; no model training or fine-tuning was performed locally [20]. The instruction asks the model to first extract domain-specific terms from the text as a JSON list, and then to produce a short, beginner-level (CEFR B1) explanation for each term. The full prompt template is given in the Appendix.

E. Evaluation Metrics

ASR layer. Speech recognition performance is evaluated with two standard, reference-based error metrics: Word Error Rate (WER) and Character Error Rate (CER). For a reference of N words,

$$\mathrm{WER} = \frac{S + D + I}{N}, \tag{1}$$

where S, D, and I are the numbers of substitutions, deletions, and insertions, respectively; CER is defined analogously at the

TABLE II: Evaluation protocol across languages.

Metric	TR	EN
Term context (R1/R0)	✓	✓
Explanation accuracy (D1/D0)	✓	✓
CEFR difficulty ($B1^{+}$)	–	✓
BLEU similarity	–	✓
Latency	✓	✓

character level over N_c reference characters. These metrics were chosen because they are standard, reference-based, and comparable across systems; CER, in addition, provides a more stable measure than word-level errors under the morphological richness of Turkish. Both metrics are computed on raw and normalized outputs. Normalization includes lowercasing, punctuation removal, and whitespace cleanup, in order to reduce comparison errors caused by output formatting. To assess practical efficiency, two further metrics are used: inference latency and the Real-Time Factor (RTF),

$$\mathrm{RTF} = \frac{t_{\mathrm{proc}}}{t_{\mathrm{audio}}}, \tag{2}$$

i.e., processing time divided by audio duration ($\mathrm{RTF} < 1$ indicates faster-than-real-time processing).

Semantic layer. The semantic outputs are evaluated against reference dictionaries using a binary labeling scheme: the Turkish Language Association (TDK) dictionary for Turkish [22] and the Oxford Dictionary for English [21]. Two labels are used: the *R* label expresses whether the extracted terminological concept is consistent with the text context, and the *D* label expresses whether the generated explanation is correct with respect to the dictionary references. From these counts we report the rates

$$\mathrm{R1\%} = \frac{R1}{R1 + R0}, \qquad \mathrm{D1\%} = \frac{D1}{D1 + D0}. \tag{3}$$

For the English dataset, two auxiliary metrics are additionally used: the CEFR-based *B1+ Ratio* and *BLEU* [19]. The B1+ Ratio is the proportion of words at CEFR level B1 or above to the total number of words in an explanation,

$$\text{B1+ Ratio} = \frac{|\{w : \mathrm{CEFR}(w) \geq \mathrm{B1}\}|}{|\{w\}|}, \tag{4}$$

so that a lower value indicates more beginner-friendly explanations. The reason these two metrics are applied only to English is an asymmetry in the underlying linguistic resources: the B1+ Ratio relies on standardized CEFR English word lists, for which there is no direct standardized Turkish equivalent, and BLEU requires aligned reference definitions. On the Turkish side, terms and explanations were evaluated through the R/D labels against TDK references. Table II summarizes which metrics apply to each language.

F. Experimental Setup and Reproducibility

For reproducibility, the checkpoints, decoding settings, and hardware used are as follows: Whisper `medium` / `large-v3` (OpenAI); Wav2Vec2 `facebook/wav2vec2-large-xlsr-53`; SpeechBrain with decoding using the default greedy/beam settings; and the experiments were run on CPU. Reference transcriptions were created by manually correcting the audio recordings and verified by a human. The LLMs were queried through the OpenAI API with default parameters. The R/D labels were assigned by a domain-knowledgeable annotator using the corresponding dictionary references.

TABLE III: Average transcription accuracy of the ASR models.

Model	WER (Raw)	CER (Raw)	WER (Norm)	CER (Norm)
Whisper Medium	0.058	0.021	0.021	0.015
Whisper Large-v3	0.132	0.113	0.129	0.112
Wav2Vec2[†]	1.017	0.848	0.548	0.196
SpeechBrain[†]	1.246	1.051	0.382	0.273

[†] Evaluated with their default, primarily English-oriented checkpoints; the values on Turkish data reflect out-of-the-box multilingual coverage rather than an architectural limit (discussed below).

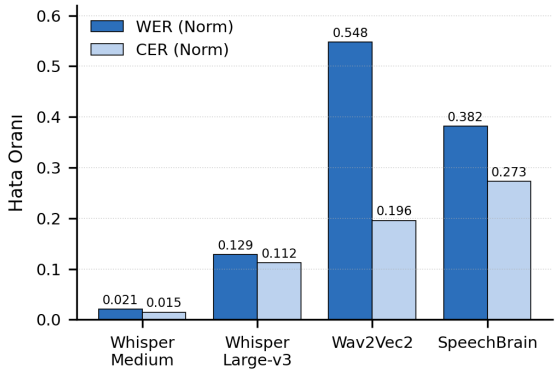


Fig. 2: Comparison of normalized WER and CER across ASR models.

TABLE IV: Computational performance of the ASR models.

Model	Mean Latency (s)	Mean RTF
Whisper Medium	136.5	0.156
Whisper Large-v3	653.4	0.703
Wav2Vec2	46.1	0.051
SpeechBrain	1795.1	2.005

IV. Experimental Results

This section presents the speech recognition and semantic analysis results. Tests were conducted separately for the Turkish and English datasets, and the effects of the components on accuracy and computational efficiency were compared.

When the results in Table III and Fig. 2 are examined, Whisper-based models clearly achieve substantially lower error rates than the other systems. In particular, the normalized WER and CER values are the primary metrics that directly reflect transcription accuracy; lower error rates make the text passed to the semantic analysis stage more reliable. As shown in Table IV, the Wav2Vec2 model, despite having lower computational cost, cannot provide sufficient transcription accuracy for semantic analysis due to its high error rates.

Two aspects of these results are worth clarifying. First, the models are compared in their off-the-shelf configura-

TABLE V: Semantic analysis results (TR and EN). R1% = R1/(R1+R0), D1% = D1/(D1+D0). $B1^+$ and BLEU are defined only for EN (see Sec. III-E).

Model	Lang.	Total Terms	R1/R0 (R1%)	D1/D0 (D1%)	$B1^+$ Ratio	BLEU
GPT-4.1	EN	105	101/4 (96.2%)	104/1 (99.0%)	0.074	0.61
GPT-4o-mini	EN	56	51/5 (91.1%)	54/2 (96.4%)	0.191	0.57
GPT-4.1	TR	78	75/3 (96.2%)	77/1 (98.7%)	–	–
GPT-4o-mini	TR	36	33/3 (91.7%)	34/2 (94.4%)	–	–

TABLE VI: Processing latency of the LLMs on the Turkish dataset.

Model	Mean (s)	Std	Min (s)	Max (s)
GPT-4.1	17.79	5.40	10.99	34.44
GPT-4o-mini	12.53	1.77	9.16	14.85

Latency statistics (min/max): GPT-4.1 10.99/34.44 s; GPT-4o-mini 9.16/14.85 s.

tions: Whisper is pre-trained on large-scale multilingual data, whereas the Wav2Vec2 and SpeechBrain checkpoints used here are primarily English-oriented. The higher error rates of the latter two on the Turkish data therefore reflect their out-of-the-box multilingual coverage rather than an inherent architectural limitation, and our results are accordingly framed as a comparison of out-of-the-box suitability for this multilingual setting; evaluating language-matched (multilingual or Turkish fine-tuned) checkpoints is a natural next step. Second, Whisper Medium attains a slightly lower normalized WER than Large-v3 (0.021 vs. 0.129), which is consistent with the larger model's known sensitivity, on long-form audio, to language detection and decoding settings. We therefore treat this as a configuration-dependent effect rather than a general ranking of the two models.

The semantic analysis results are presented in Table V. The results show that GPT-4.1 detects more terminological expressions than GPT-4o-mini. Despite this, the R1% and D1% rates remain high, indicating that the model can preserve context consistency and explanation accuracy even though it selects more terms. On the English dataset, the lower $B1^+$ value obtained with GPT-4.1 indicates that fewer advanced-level words are used in the explanations and that the definitions are more comprehensible for beginner-level users. As an additional comparison, we report the per-call processing latency of the two LLMs on the Turkish subset in Table VI. GPT-4o-mini provides lower and more stable processing time than GPT-4.1, indicating that the smaller model offers an advantage in computational efficiency.

When the results are considered overall, different model combinations may be preferred for different application scenarios. For accuracy-oriented applications that require higher semantic coverage and terminological richness, Whisper Medium transcription with GPT-4.1 semantic analysis offers a more suitable configuration, whereas for systems with limited computational budget or low-latency operation, the Whisper Medium with GPT-4o-mini combination can provide a more efficient solution with lower processing latency.

V. Limitations

This study is a focused benchmark, and a few points should be kept in mind when interpreting the results. (i) *Dataset size:* the dataset comprises 7 videos (∼95 minutes); the reported values are averages over this set, and larger-scale evaluation with significance testing is a natural extension. (ii) *Scope of the ASR comparison:* the systems are evaluated with their default checkpoints, so the results characterize out-of-the-box behavior in this multilingual setting rather than each architecture's best attainable accuracy. (iii) *Semantic evaluation:* the R/D and B1+ measures involve some human judgment, and a multi-annotator protocol with agreement analysis would further strengthen these results. (iv) *Language coverage of metrics:* BLEU and B1+ are reported for English only, for the reasons given in Sec. III-E. (v) *API-based LLMs:* the LLMs are accessed through a hosted API, so exact outputs may evolve over time. These points define the scope of the present evaluation and motivate the extensions outlined in Sec. VI.

VI. Conclusion and Future Work

In this study, we presented a two-layer experimental evaluation of a speech-based learning support system aimed at improving the comprehensibility of multilingual educational content for beginner-level users. In the first layer, different ASR systems were compared in terms of transcription accuracy (WER, CER) and computational efficiency (latency, RTF); Whisper-based models were shown to provide more reliable transcriptions for semantic processing through lower error rates. In the second layer, LLM-based terminology extraction and context-aware explanation generation were performed on the transcriptions; GPT-4.1 offered higher terminology coverage and explanation accuracy, while GPT-4o-mini provided lower and more stable latency.

The findings show that speech recognition accuracy directly affects the reliability of semantic analysis outputs, and that jointly considering ASR and LLM components is critical in the design of speech-based educational systems. In future work we aim to address: (a) evaluating language-matched (multilingual or Turkish fine-tuned) checkpoints for Wav2Vec2/SpeechBrain; (b) a *cross-layer* analysis in which ASR outputs of different quality are fed to the same LLM, directly supporting the central claim; (c) intermediate modules between ASR and the LLM, such as error correction / confidence-based filtering and ASR-aware prompting; (d) the effect of prompt-engineering, chain-of-thought (CoT), and retrieval-augmented generation (RAG) strategies on terminology extraction; and (e) a larger dataset, statistical tests,

multi-annotator IAA measurement, and symmetric metrics for both languages. We further plan to transform the system into an adaptive learning structure that adjusts to the user's level of prior knowledge, presenting only the terminological explanations that are needed based on user interactions and past performance.

APPENDIX A
PROMPT TEMPLATE

The following instruction was used (held fixed across all runs); `{LANG}` and `{TRANSCRIPT}` are substituted at call time. The exact wording is available in the shared repository.

```
You are a terminology assistant for {LANG}
educational content. Step 1: From the
transcript below, extract domain-specific
terms (including multi-word terms). Step 2:
For each term, write one short explanation
at CEFR B1 level that a beginner can
understand. Use only information consistent
with the transcript context. Return strictly
a JSON array of objects with fields "term"
and "explanation". Transcript: {TRANSCRIPT}
```

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